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22F-3277
LAB 11
03/11/2023

LAB 11

(Total Marks 15+3=18)

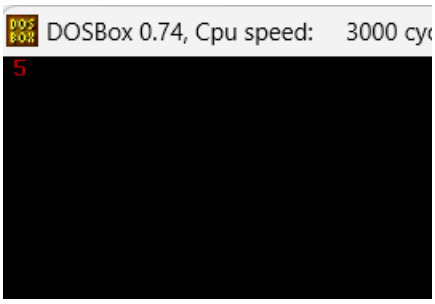
Task 1: Write the program that count the number of spaces in given string using SCAS instruction and display the counter on screen

- Determine the length of a string using the null terminator.
- Only use SCAS instruction
- Message: db 'Ass em bly lan gua ge', 0 counter: 5
- Use your full name in message and put space after two alphabetics.

```
org 0x0100

jmp start

message: db 'Mu ha mm ad Ta ha', 0

clrscr:
        push es
        push ax
        push cx
        push di

        mov ax,0xb800
        mov es,ax
        mov cx,2000
        mov ax,0x0720

        cld
        rep stosw

        pop di
        pop cx
        pop ax
        pop es
```

```
        ret

countspaces:
        push bp
        mov bp,sp
        push ax
        push es
        push cx
        push di

;;;;;;;
;calculating length

        les di, [bp+4]     ;es:di to string, es point to segment, di point to string
        mov cx, 0xffff
        xor al,al
        repne scasb

        mov ax, 0xffff
        sub ax, cx
        dec ax                  ;minus null value from string length
        mov cx, ax              ;moves length of string in ax
;;;;;;;;

        les di, [bp+4] ;es:di is again set correct
        mov al, ' '
        mov bx, 0               ;bx counts spaces

next:
        repne scasb
        jne printCount
        inc bx
        jmp next

        call printCount

        pop di
        pop cx
        pop es
        pop ax
        pop bp
        ret 4

printCount:
        push bx
        push ax
        push di
        push es
```

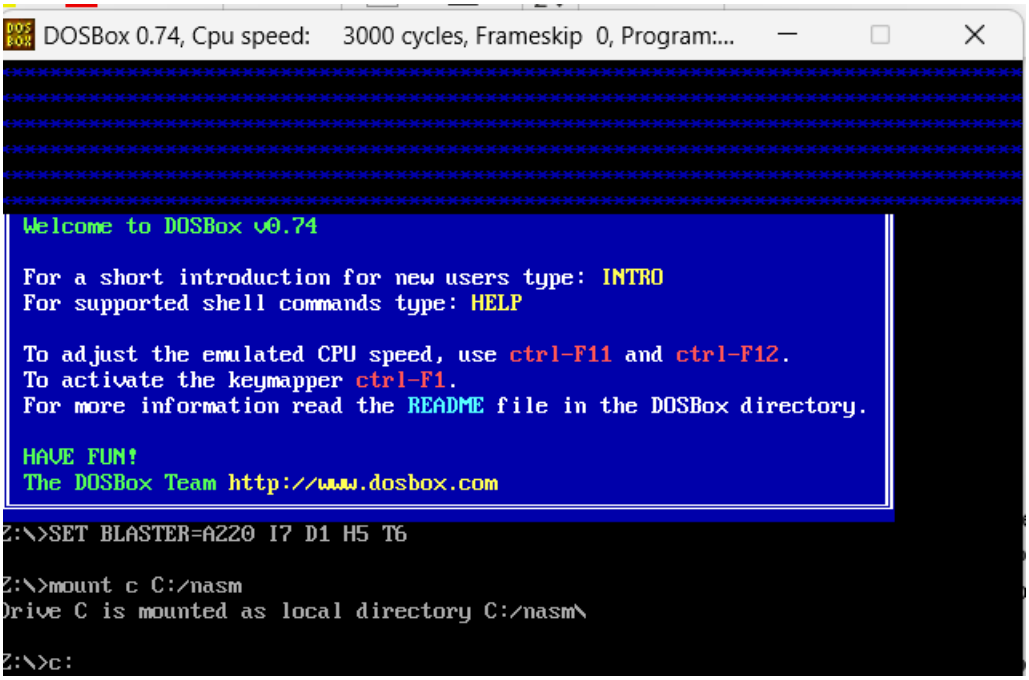
```
        mov ax, 0xb800
        mov es, ax
        mov di,160

        add bl, 0x30
        mov bh, 4h                  ;red color
        mov ax,bx
        add di, 2
        mov [es:di], ax ; print char on screen

        pop di
        pop di
        pop ax
        pop bx
        ret

start:
        call clrscr

        push ds
        mov ax, message
        push ax

        call countspaces

        mov ax,0x4c00
        int 21h
```



Task 2: Write a code for scroll down behavior of the screen. This will require an input of how many numbers of lines being needed to be scrolled up as a parameter and then you will use "Movs" to scroll down.

```
; scroll down the screen
[org 0x0100]
 jmp start
; subroutine to scrolls down the screen
; take the number of lines to scroll as parameter
scrolldown:
        push bp
        mov bp,sp
        push ax
        push cx
        push si
        push di
        push es
        push ds

        mov ax, 80 ; load chars per row in ax
        mul byte [bp+4] ; calculate source position
        push ax ; save position for later use
        shl ax, 1 ; convert to byte offset
        mov si, 3998 ; last location on the screen
        sub si, ax ; load source position in si

        shr ax,1          ;;;div ax by 2 so now we know the number of words

        mov cx, 2000 ; number of screen locations
        sub cx, ax ; count of words to move
        mov ax, 0xb800
        mov es, ax ; point es to video base
        mov ds, ax ; point ds to video base
        mov di, 3998 ; point di to lower right column

        std ; set auto decrement mode
        rep movsw ; scroll up
        mov ax, 0x12a ;;;put star above
        pop cx ; count of positions to clear
        rep stosw ; clear the scrolled space


    pop ds
        pop es
        pop di
        pop si
```

```
        pop cx
        pop ax
        pop bp
        ret 2

start:
        mov ax,6
        push ax ; push number of lines to scroll
        call scrolldown ; call scroll down subroutine
        mov ax, 0x4c00 ; terminate program
        int 0x21
```



Task3: Write a program for substring search (that searches for a substring within a given string) The program should return the 1 if the substring is found, or 0 if the substring is not found.

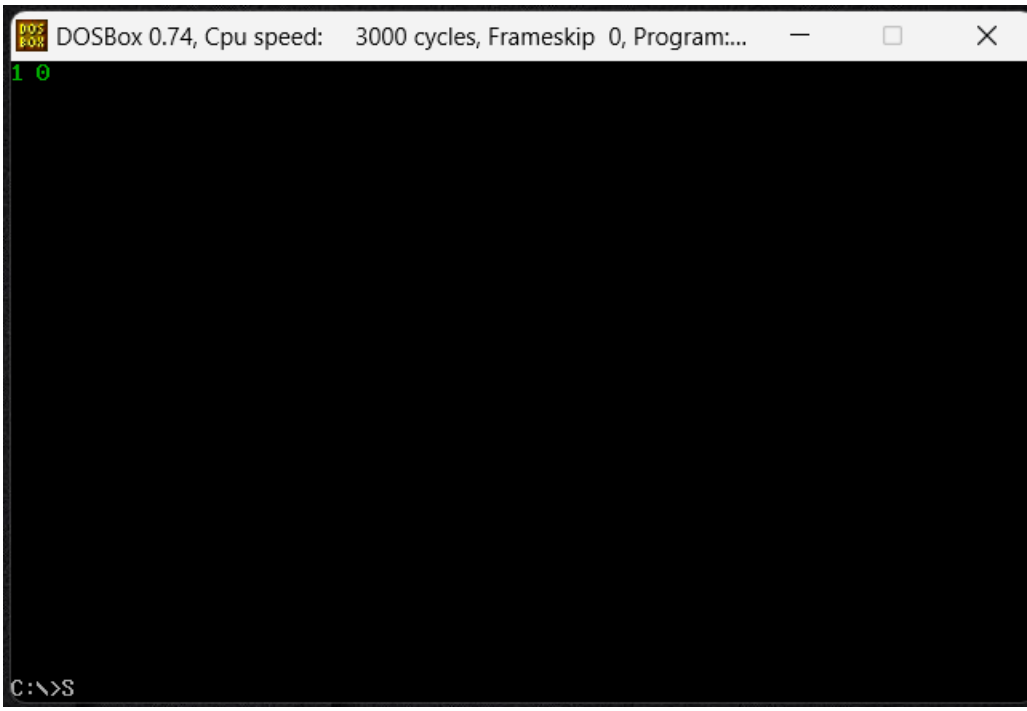
- Use Cmps instruction
- Use ax register for returning value
- i.e message string: db 'hello world'
- search: db 'world'
- Use your full name in message string and once find first name in message string and then search wrong name.

```
; find substring from string
[org 0x0100]
jmp start
string: db 'Muhammad Taha', 0
substring1: db 'Taha', 0
substring2: db 'xyz', 0

clrscr:
        push es
        push ax
        push cx
        push di

        mov ax,0xb800
        mov es,ax
        mov cx,2000
        mov ax,0x0720

        cld
        rep stosw

        pop di
        pop cx
        pop ax
        pop es
        ret

strlen:
        push bp
        push es
        push cx
        push di
        push si
```

```
        mov cx, 0xffff ; load maximum number in cx
        xor al, al ; load a zero in al
        repne scasb ; find zero in the string
        mov ax, 0xffff ; load maximum number in ax
        sub ax, cx ; find change in cx
        dec ax ; exclude null from length

        pop si
        pop di
        pop cx
        pop es
        pop bp
        ret

findSubstring:
        push bp
        mov bp,sp
        push cx
        push si
        push di
        push es

        mov es, [bp+8]  ;points to ds

        mov di, [bp+4]  ;now es:di points to substring to calculate length
        call strlen             ;returns lenght of substring
        mov bx, ax              ;lenght of substring


        mov di, [bp+6]  ;point es:di to string
        call strlen             ;returns length of string
        mov cx, ax              ;lenght of string

        mov si,[bp+4]   ;point ds:si to substring

        cmp bx, cx              ;if substring is greater
        jg notfound

        ;es:di points to string
        ;ds:si points to substring to find in string

l1:
        repe cmpsb
        cmp cx,0
        je exit
        mov si, [bp+4]
        jne l1
```

```
exit:

        add bx, [bp+4]   ;adding substring index to the length of substring and tha comparing if it is equal to current si value than it means that the substring was found

        cmp si, bx
        je found
        jne notfound

        found:
        mov ax,1
        jmp return

        notfound:
        mov ax,0

return:

        pop es
        pop di
        pop si
        pop cx
        pop bp
        ret 6

printAX:
        push di
        push ax
        push bx
        push es


        mov bx, 0xb800
        mov es,bx

        add al, 0x30
        mov ah, 02
        mov word [es:di],ax

        pop es
        pop bx
        pop ax
        pop di
        ret 4
start:

        call clrscr
```

```
push ds
mov ax, string
push ax
mov ax, substring1
push ax

call findSubstring

push ax
mov di, 160       ;push di value
push di
call printAX

push ds
mov ax, string
push ax
mov ax, substring2
push ax

call findSubstring ;returns 1 or 0 in AX

push ax
mov di, 164       ;push di value
push di
call printAX

mov ax, 0x4c00
int 0x21
```



Best of luck